

# UNIVERSITY COLLEGE LONDON

## EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : **COMP0085**

ASSESSMENT : **COMP0085A7PA**  
PATTERN

MODULE NAME : **Approximate Inference and Learning in  
Probabilistic Models**

LEVEL: : **Postgraduate**

DATE : **15 May 2019**

TIME : **14:30**

TIME ALLOWED : **2 hrs 30 mins**

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

**Year**  
**2018-19**

**EXAMINATION PAPER CANNOT BE REMOVED FROM THE EXAM HALL. PLACE EXAM  
PAPER AND ALL COMPLETED SCRIPTS INSIDE THE EXAMINATION ENVELOPE**

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|-------------------------------------|---|
| <b>Hall Instructions</b>            | . |
| <b>Standard Calculators</b>         | Y |
| <b>Non-Standard<br/>Calculators</b> | N |
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**TURN OVER**

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**Approximate Inference and Learning in  
Probabilistic Models, COMP0085 (A7P)**  
Main Summer Examination Period, 2018/19

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This examination contains SIX questions, of which you should answer FIVE.

Each question is worth 20 marks. If you have time, you may attempt all 6 questions, but only your 5 best scores will count towards your final mark. Make sure you manage your time carefully, read through ALL questions first before attempting any, and don't spend too much time on any one question. Calculators are allowed, but are unlikely to be useful.

**Always provide justification and show any intermediate work for your answers.  
A correct but unsupported answer may not receive any marks.**

In the following, bold symbols represent vectors. When asked to provide algorithms, you may provide equations or use MATLAB/Octave notation for matrix algebra. You may assume the availability of standard matrix functions such as `inv`, `pinv`, `eig` or `svd`. You may **not** assume statistical functions, including `mean` and `cov`. If you choose to use MATLAB notation, you will not lose marks for syntactic errors, as long as your intention is clear and correct.

You might find it useful to consult the following table of standard distributions:

| Name                | Domain                 | Symbol                                                  | Density or Probability fn                                                                                                                     |
|---------------------|------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Gaussian (Normal)   | $\mathbb{R}$           | $x \sim \mathcal{N}(\mu, \sigma^2)$                     | $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$                                                                   |
| Multivariate Normal | $\mathbb{R}^D$         | $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ | $ 2\pi\Sigma ^{-1/2} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}$                             |
| Bernoulli           | $\{0, 1\}$             | $x \sim \text{Bernoulli}(p)$                            | $p^x (1-p)^{1-x}$                                                                                                                             |
| Binomial            | $\mathbb{Z}_{0-N}$     | $x \sim \text{Binom}(p)$                                | $\binom{N}{x} p^x (1-p)^{N-x}$                                                                                                                |
| Multinomial         | $[\mathbb{Z}_{0-N}]^D$ | $\mathbf{x} \sim \text{Multinom}(\mathbf{p})$           | $\frac{N!}{x_1! x_2! \dots x_D!} \prod_{d=1}^D p_d^{x_d}$                                                                                     |
| Poisson             | $\mathbb{Z}_{0+}$      | $x \sim \text{Poisson}(\mu)$                            | $\mu^x e^{-\mu} / x!$                                                                                                                         |
| Beta                | $[0, 1]$               | $x \sim \text{Beta}(\alpha, \beta)$                     | $\frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$                                                                                     |
| Exponential         | $\mathbb{R}_+$         | $x \sim \text{Exp}(\lambda)$                            | $\lambda e^{-\lambda x}$                                                                                                                      |
| Gamma               | $\mathbb{R}_+$         | $x \sim \text{Gamma}(\alpha, \beta)$                    | $\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$                                                                               |
| Student-t           | $\mathbb{R}$           | $x \sim \text{Student-t}(\alpha)$                       | $\frac{\Gamma(\frac{\alpha+1}{2})}{\sqrt{\pi\alpha}\Gamma(\frac{\alpha}{2})} \left(1 + \frac{x^2}{\alpha}\right)^{-\frac{\alpha+1}{2}}$       |
| Inverse-Wishart     | $\mathbb{S}_+^p$       | $X \sim \text{InvWish}(\Psi, \nu)$                      | $\frac{ \Psi ^{\frac{\nu}{2}}}{2^{\frac{\nu p}{2}} \Gamma_p(\frac{\nu}{2})}  X ^{-\frac{\nu+p+1}{2}} e^{-\frac{1}{2} \text{Tr}[\Psi X^{-1}]}$ |
| Dirichlet           | $[0, 1]^D$             | $\mathbf{x} \sim \text{Dirichlet}(\boldsymbol{\alpha})$ | $\frac{\Gamma(\sum_{d=1}^D \alpha_d)}{\prod_{d=1}^D \Gamma(\alpha_d)} \prod_{d=1}^D x_d^{\alpha_d-1}$                                         |

## 1. RBM unrolling

Recall that a restricted Boltzmann machine (RBM) is an undirected graphical model on observed and latent binary variables with pairwise factors, such that each edge connects an observed and a latent node. Let  $v_i, i = 1 \dots V$  be the observed or *visible* variables, collected into a binary vector  $\mathbf{v} \in \{0, 1\}^V$  and  $h_j, j = 1 \dots H$  be the latent or *hidden* variables collected into  $\mathbf{h} \in \{0, 1\}^H$ .

Then an RBM with weights  $W_{ij}$  (collected into a matrix  $W$ ) implies the joint distribution:

$$p(\mathbf{v}, \mathbf{h}) \propto e^{\sum_{i=1}^V \sum_{j=1}^H W_{ij} v_i h_j} = e^{\mathbf{v}^T W \mathbf{h}}.$$

- Find the conditional distributions  $P_{vh} = P(\mathbf{v}|\mathbf{h})$  and  $P_{hv} = P(\mathbf{h}|\mathbf{v})$ , demonstrating that they can be evaluated and sampled tractably. [4 marks]
- Why is tractability of the conditionals not sufficient to make maximum likelihood learning of the weights  $W_{ij}$  tractable? [2 marks]
- Derive the Boltzmann machine learning rule as it applies to the RBM. Describe how to evaluate both the clamped and unclamped distributions as efficiently as possible using Gibbs sampling where needed. [4 marks]

The restricted structure creates an interesting link between the RBM and a directed model. Consider a *directed* graph or DAG over a sequence of vector binary variables  $\mathbf{v}, \mathbf{h}^{(0)}, \mathbf{v}^{(1)}, \mathbf{h}^{(1)}, \mathbf{v}^{(2)}, \mathbf{h}^{(2)}, \dots, \mathbf{v}^{(N)}, \mathbf{h}^{(N)}$  (with  $\mathbf{h}^{(n)} \in \{0, 1\}^H$  and  $\mathbf{v}^{(n)} \in \{0, 1\}^V$  for all  $n$ ), such that:

$$\begin{aligned} \mathbf{h}^{(N)} &\sim P_h \\ \mathbf{v}^{(n)}|\mathbf{h}^{(n)} &\sim P_{vh} & n = N \dots 1 \\ \mathbf{h}^{(n-1)}|\mathbf{v}^{(n)} &\sim P_{hv} & n = N \dots 1 \\ \mathbf{v}|\mathbf{h}^{(0)} &\sim P_{vh} \end{aligned}$$

where  $P_h$  is a known fixed distribution and  $P_{hv}$  and  $P_{vh}$  are the  $W$ -dependent RBM conditional distributions you found in part (a).

- As the number of layers  $N \rightarrow \infty$  how does the marginal  $P(\mathbf{v})$  of the output of this DAG relate to the RBM distribution with the same weight matrix  $W$ ? Explain. [2 marks]
- Again working in the limit  $N \rightarrow \infty$ , find the inferential distribution  $P(\mathbf{h}^{(0)}|\mathbf{v})$  in terms of  $W$ . [4 marks]
- Consider a sampled partial expectation-maximisation approach to learn  $W$  in the DAG. First, for each observed binary vector  $\mathbf{v}_k$ , we sample  $\mathbf{h}_k^{(0)}$  from  $P(\mathbf{h}^{(0)}|\mathbf{v}_k)$  (as found in (e)) using the current value of  $W$ . We then optimise  $W$  to maximise  $\left\langle \sum_k \log P(\mathbf{v}_k|\mathbf{h}_k^{(0)}) \right\rangle$ . Show that if the  $\mathbf{v}_k$  are drawn from an RBM with weight matrix  $W^*$  then  $W^*$  is an asymptotic stationary point of this algorithm, given infinite data and samples. [4 marks]

## 2. GP point process models

A temporal point process generates discrete events in time. The simplest is the Poisson process, under which the event distribution depends only on an intensity function  $\lambda(t)$ . Specifically, the probability of observing events at times  $\{t_1 \dots t_N\}$  within an interval  $[0, T]$  is given by

$$P(t_1 \dots t_N | \lambda) = \left( \prod_{n=1}^N \lambda(t_n) \right) e^{-\int_0^T \lambda(t) dt}.$$

Consider a doubly-stochastic model in which  $\lambda(t)$  is a random function, drawn from a transformed Gaussian process with mean function  $\mu$  and covariance function  $\kappa$ :

$$\lambda(t) = e^{x(t)}; \quad x(t) \sim \mathcal{GP}(\mu(\cdot), \kappa(\cdot, \cdot))$$

We will derive an inducing-point based variational algorithm to estimate  $P(x(\cdot) | t_1 \dots t_N)$ .

Define inducing point times  $\tau_1 \dots \tau_M$ , collected into a vector  $\tau$ , and values  $u_1 \dots u_M$  collected into  $\mathbf{u}$ . Let the variational distribution on  $\mathbf{u}$  be  $q(\mathbf{u}) = \mathcal{N}(\mathbf{u}; \mathbf{m}, S)$ , and introduce the vector and matrix notation  $\boldsymbol{\mu}_\tau = [\mu(\tau_m)]$ ,  $K_{\tau, \tau} = [\kappa(\tau_m, \tau_{m'})]$  and  $\mathbf{k}_\tau(t) = [\kappa(\tau_m, t)]$ .

- (a) Using the standard definition of  $q(x(\cdot), \mathbf{u})$ , show that the augmented variational KL divergence can be written:

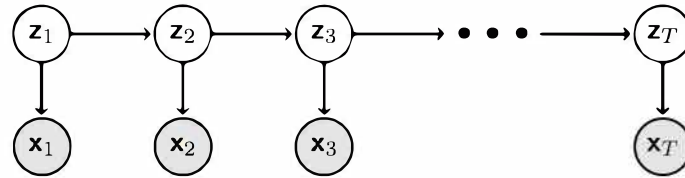
$$\begin{aligned} \text{KL}[q(x, \mathbf{u}) || P(x, \mathbf{u} | t_1 \dots t_N)] = & - \left\langle \log P(t_1 \dots t_N | x) \right\rangle_{P(x | \mathbf{u})} \Big|_{q(\mathbf{u})} \\ & + \text{KL}[\mathcal{N}(\mathbf{u}; \mathbf{m}, S) || \mathcal{N}(\mathbf{u}; \boldsymbol{\mu}_\tau, K_{\tau, \tau})] + \text{constants}. \end{aligned}$$

[6 marks]

- (b) Express  $P(x | \mathbf{u})$  in terms of model and variational parameters. [4 marks]
- (c) Consider the first (likelihood) term in the expression of part (a). Evaluate the expectations in closed form to show that this term can be computed exactly (apart from an integral over time, which must be evaluated numerically). You may wish to use the fact that  $\langle e^z \rangle_{\mathcal{N}(z; \mu, \sigma^2)} = e^{\mu + \sigma^2/2}$ . [8 marks]
- (d) The variational parameters  $\mathbf{m}$ ,  $S$  can now be found by gradient descent, differentiating both this term and the KL divergence to the prior. We can also treat the locations  $\tau_m$  as variational parameters. Considering the influence of both the predictive mean and the predictive uncertainty in your expression for the KL divergence, discuss the likely optimal distribution of inducing point locations. [2 marks]

### 3. Particle Filtering

Consider a general state-space model:



in which the conditional dependencies may be non-linear or non-Gaussian.

Suppose that we have a way of generating true samples from  $P(\mathbf{z}_1|\mathbf{x}_1)$ . We use the “condensation filter” to obtain importance-weighted samples from  $P(\mathbf{z}_2|\mathbf{x}_1, \mathbf{x}_2)$  (strictly, we should see them as samples from the joint posterior  $P(\mathbf{z}_1, \mathbf{z}_2|\mathbf{x}_1, \mathbf{x}_2)$ , but we will analyse the marginal posterior on  $\mathbf{z}_2$ ).

We are interested in analysing the behaviour of the condensation filter, by linking the distribution of the samples and importance weights to the marginal and conditional distributions of the model. Express all of your answers below in the simplest form you can.

First, for each sample  $\mathbf{z}_1^s \sim P(\mathbf{z}_1|\mathbf{x}_1)$ , we draw  $\mathbf{z}_2^s \sim P(\mathbf{z}_2|\mathbf{z}_1^s)$ .

- (a) What is the marginal distribution of these samples? Justify your answer.

[4 marks]

Next, we calculate the importance weights  $w_2^s \propto P(\mathbf{x}_2|\mathbf{z}_2^s)$ .

- (b) What is the expected value of the unnormalised weights (and therefore the expected value of the normaliser)?

[4 marks]

Assume for simplicity that the normaliser takes on this expected value exactly (i.e., it does not depend on the actual particle samples  $\mathbf{z}_2^s$ ).

- (c) Find an expression for the variance of the importance weights.

[4 marks]

Consider the following two probabilities.  $P_2$  is the probability that *two* independent identical chains, both of which generated the observed  $\mathbf{x}_1$  at time step 1 will then also both generate the observed  $\mathbf{x}_2$  at time step 2.  $P_1$  is the probability that a *single* chain with two outputs at time step 2 which are iid when conditioned on  $\mathbf{z}_2$  will produce two copies of the observed  $\mathbf{x}_2$ , having produced (only one copy of) the observed  $\mathbf{x}_1$  at time step 1.

- (d) Show that your computed variance is equal to  $\frac{P_1}{P_2} - 1$ .

[4 marks]

- (e) Does this mean the variance will go up or down as the observed value of  $\mathbf{x}_2$  becomes more or less likely given  $\mathbf{x}_1$ ?

[4 marks]

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#### 4. Belief Propagation

Consider a Markov random field (MRF) or undirected graphical model with pairwise potentials:

$$P(x_1 \dots x_n) \propto \prod_{(ij) \in \mathcal{E}} f(x_i, x_j)$$

where  $\mathcal{E}$  is an edgese~~t~~.

- (a) Define the Bethe free energy for this model. *[4 marks]*
- (b) What approximations are made when we replace optimisation of the true variational free-energy for such a random field with optimisation of the Bethe free energy? *[4 marks]*
- (c) Under what conditions on the graphical model is the Bethe free energy exact? *[2 marks]*
- (d) If belief propagation (BP) converges to a stable fixed point can we conclude that the Bethe free energy is exact? *[3 marks]*
- (e) If BP converges, we will have obtained stable estimates of locally-consistent pairwise and singleton beliefs  $b(x_i, x_j)$  and  $b(x_i)$ . How is the product

$$\prod_i b(x_i) \prod_{(ij) \in \mathcal{E}} \frac{b(x_i, x_j)}{b(x_i) b(x_j)}$$

related to the true joint distribution? Explain. *[4 marks]*

- (f) Consider a (spanning) subtree of the graph, defined by an edgese~~t~~  $\mathcal{E}_T \subset \mathcal{E}$ . What more can be said about the BP beliefs and the distribution

$$P_T(x_1 \dots x_n) \propto \prod_{(ij) \in \mathcal{E}_T} f(x_i, x_j) ?$$

*[3 marks]*



## 5. Modelling Crime Rates

The police have collected a dataset of crime incidence across London, and would like to model how crime rates vary across different boroughs and in time. Index the boroughs by  $i = 1, \dots, N$  and months by  $t = 1, \dots, T$ . Let  $C_{ij}$  be the number of crimes in borough  $i$  in month  $t$ .

You construct a simple model in which each borough  $i$  has an intrinsic crime rate  $r_i$  that is independent of the crime rates in other boroughs, and each month  $t$  the crime rates are amplified by a scale factor  $s_t$ . Specifically, you assume:

$$\begin{aligned} r_i &\sim \text{Gamma}(a, b) && \text{for each borough } i \\ s_t &\sim \text{Gamma}(c, d) && \text{for each month } t \\ C_{it} &\sim \text{Poisson}(r_i s_t) && \text{for each } i, t \end{aligned}$$

- Show that the posterior over the rates  $r_i$  and scales  $s_t$  when the incidence counts  $C_{it}$  are all observed can be expressed as a fully connected bipartite undirected graph, with the rates on one side, and the scales on the other. [4 marks]
- What are the conditional independencies exhibited by the posterior when the incidence counts  $C_{it}$  are observed? [4 marks]
- Derive a variational algorithm to infer the rates and scales from fully observed incidences  $C_{it}$ , assuming that

$$\{r_i\}_{i=1}^N \perp\!\!\!\perp \{s_t\}_{t=1}^T.$$

[6 marks]

- Derive an expectation propagation algorithm to infer the rates and scales. Approximate the likelihood term for  $C_{it} \sim \text{Poisson}(r_i s_t)$  by

$$L_{it} \approx r_i^{a_{it}} e^{-b_{it} r_i} s_t^{c_{it}} e^{-d_{it} s_t}$$

where  $a_{it}$ ,  $b_{it}$ ,  $c_{it}$  and  $d_{it}$  are variational parameters to be fit using EP. You may assume that the KL projection step is tractable. [6 marks]



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## 6. Tightening the variational bound with importance weights

Consider a variational autoencoder used to train a generative latent model  $p_\theta(\mathbf{x}, \mathbf{z})$  for observed vectors  $\mathbf{x}$  depending on latents  $\mathbf{z}$ , with parameters  $\theta$ . Let  $\mathbf{f}_\phi(\mathbf{x})$  be the encoder or recognition network with parameters  $\phi$ , the outputs of which represent the parameters of a variational posterior distribution  $q(\mathbf{z}; \mathbf{f}_\phi(\mathbf{x}))$ . We assume that it is possible to evaluate  $q$ , and to sample from it by reparametrisation. That is, given  $\epsilon^s \sim \mathcal{N}(0, 1)$  there is a known deterministic function  $\mathbf{h}$  such that  $\mathbf{z}^s = \mathbf{h}(\epsilon^s, \mathbf{f}_\phi(\mathbf{x}))$  is a sample from  $q$ .

- (a) Write the variational free energy in terms of the functions  $p_\theta$ ,  $q$ ,  $\mathbf{f}_\phi$ ,  $\mathbf{h}$  and samples  $\epsilon^s$  (for  $s = 1 \dots S$ ). [2 marks]

When working with samples from the posterior distribution, it is possible to develop a different and tighter bound than the standard free energy. Consider the expression:

$$\mathcal{F}_S = \left\langle \log \frac{1}{S} \sum_{s=1}^S \frac{p_\theta(\mathbf{x}, \mathbf{z}^s)}{q(\mathbf{z}^s; \mathbf{f}_\phi(\mathbf{x}))} \right\rangle$$

where the expectation is with respect to all the  $\mathbf{z}^s \stackrel{\text{iid}}{\sim} q(\cdot; \mathbf{f}_\phi(\mathbf{x}))$ .

- (b) Using Jensen's inequality, show that  $\mathcal{F}_S \leq \log p_\theta(\mathbf{x})$ . [6 marks]
- (c) Also show that if  $S < S'$  then  $\mathcal{F}_S \leq \mathcal{F}_{S'}$ . [Hint: replace the average of  $S'$  terms in the definition of  $\mathcal{F}_{S'}$  by an expectation of a sum of  $S$  of those terms drawn uniformly at random, and use Jensen.] [6 marks]
- (d) Describe an autoencoder framework to optimise  $\mathcal{F}_s$  in place of the standard variational free energy. Give the estimates of derivatives with respect to  $\theta$  and  $\phi$ . [6 marks]